

Medallion architecture plan

Bronze:

Files:

1. Airport flight data (CSV files, EuroControl, <https://ansperformance.eu/data/>)
airport_flights_(2020-2025)_raw.csv (5 files)
2. Airport locations (mapping, <https://openflights.org/>)
airport_locations_raw.csv
3. Geographical table (city, country, continent)
geography_raw.csv

Information:

Scope: European airports, 5 years (2021-2025). Add columns: source name, file/API name, load timestamp or run id. Source data quality checks.

Silver:

Tables:

1. Airport flight table: Combining airport yearly data (concat)
stg_airport_flights
2. Geographical table (city, country, continent)
stg_geography
3. Airport locations (coordinates)
stg_airport_locations
4. Airport geographics (Joined tables geographics and airport locations)
airport_geography_combined
5. Weather data (Openmeteo weather: <https://open-meteo.com/>) (airport location information: from cleaned mapping dataset)
weather_hourly_raw.csv -> stg_weather_hourly. Timely aggregation (daily, hourly) -> Weather data summarization stg_weather_daily

Validations:

(from 1,4,5 joined tables)

6. validated_rows

7. rejected_rows

Information:

Silver must contain cleaned, standardized (columns), typed and validated data. This is where column names, dates, numeric values, category names and join keys should become reliable. Quality checks: (duplicates, null values, valid values, valid range, summary for rejections reasons).

Gold:

Tables:

fact_flights (from validated_rows)

dim_date

dim_geography

Information:

Grain: flights.

PowerBI

Analyze European airport flight traffic from 2021–2025 and evaluate how weather conditions relate to daily flight volumes across airports, countries, and seasons.